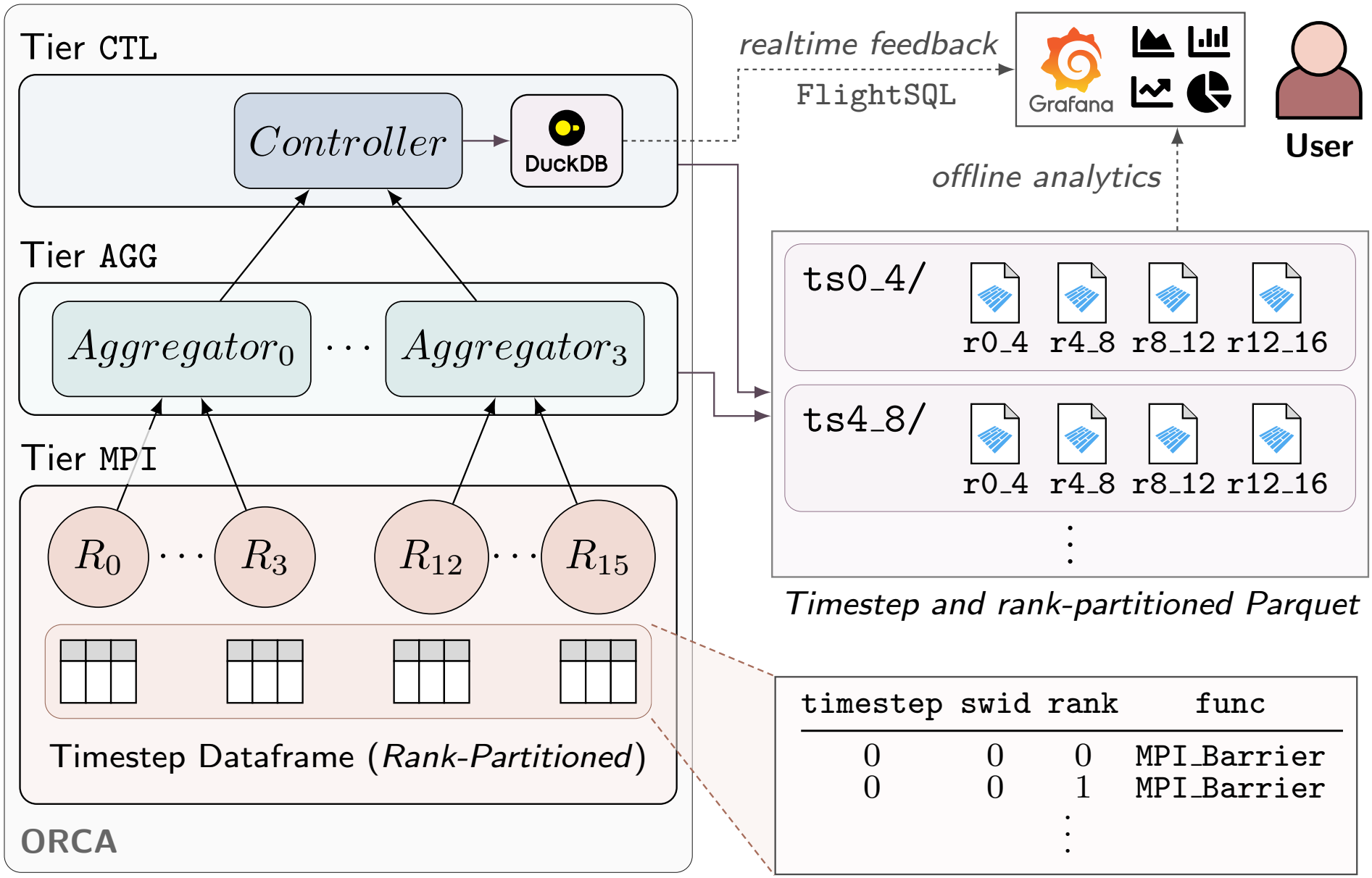

Tier CTL
Controller
DuckDB
Tier AGG
$Aggregator_0$ ⋯ $Aggregator_3$
Tier MPI
R_0 ⋯ R_3 R_{12} ⋯ R_{15}
Timestep Dataframe (*Rank-Partitioned*)
ORCA
realtime feedback
FlightSQL
Grafana
User
offline analytics
ts0_4/
r0_4 r4_8 r8_12 r12_16
ts4_8/
r0_4 r4_8 r8_12 r12_16
Timestep and rank-partitioned Parquet
timestep swid rank func
0 0 0 MPI_Barrier
0 0 1 MPI_Barrier